HEINZ NIXDORF INSTITUT
UNIVERSITÄT PADERBORN

Knowledge Graphs: Construction, Reasoning, and Retrieval-Augmented Generation

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Tutor: Asep Fajar Firmansyah

DICE
Data Science Group
Paderborn University

Project Group - WiSe 2026
July 27, 2026

About DICE

What does the DICE group do?

- Data Integration
- Data Storage and Querying
- Interpretable Data Processing
- Machine Learning
- Natural Language Processing

For more information: https://dice-research.org/groups/

Section 1

PG Organization

PG Structure and Deliverables

How is this project group structured?

- **Semesters:** WiSe 26/27 and SoSE 27
- **Registration:** You need to register your self before 31.10.2026.
- **During Semesters:**
  - Regular Group Meetings – (Usually) Every Two Weeks
  - Weekly Report
- **Semester-end Requirements:**
  - Individual Report, Group Report, and Group Presentation
  - Grading and Feedback
- **Final Submission:** Source Code and Documentation

Expectations

What do we expect from you?

→ Be a valuable member of your team!

- Manage your project
- Write code(!) and commit it
- Communicate with team members
- Support each other where possible
- …

→ We won't keep students if they are not participating in their team

PG Schedule

- **Kickoff Meeting:** 15.10.2026, FU.519
- **Regular Meetings:** Data DBD, FU.519
- **Registration:** Studienleistung and Register for Exam before the deadline—or leave the PG!
- **End WiSe 26/27:**
  - Individual Report and Group Presentation
  - Grading and Feedback
- **Between WiSe 26/27 and SoSe 27:** Vacations
- **Start SoSe 27:** Resume PG Work
- **Mid-July 2027:** End of the Implementation Phase
- **End SoSe 26:**
  - Group Report and Group Presentation
  - Final Grading

Unforeseen Events

- **Vacations:**
  - Should be avoided during lecture time!
  - In very urgent cases, contact your supervisor
- **Project Issues:**
  - Problems within the team
  - Hardware issues
  - Personal issues

  $\rightarrow$ Get in touch with your supervisor
- **Illness:**
  - Get a Doctor's Certificate!
  - If you can foresee it, inform your teammates

Section 2

PG Objectives Recap

Knowledge Graphs

Introduction

A Knowledge Graph (KG) represents entities as nodes and their relationships as edges.

- Knowledge Graphs Everywhere



- Many open knowledge graphs



- Powering AI



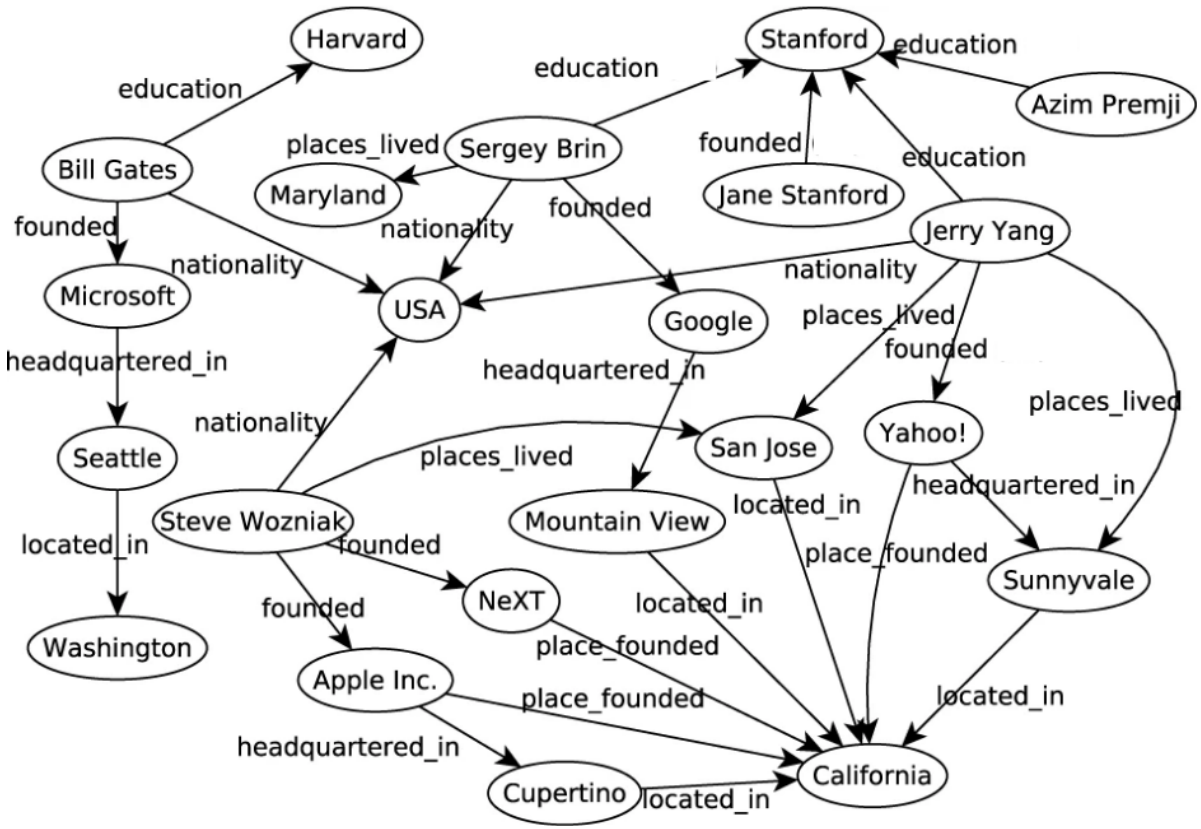
- Applied across diverse domains



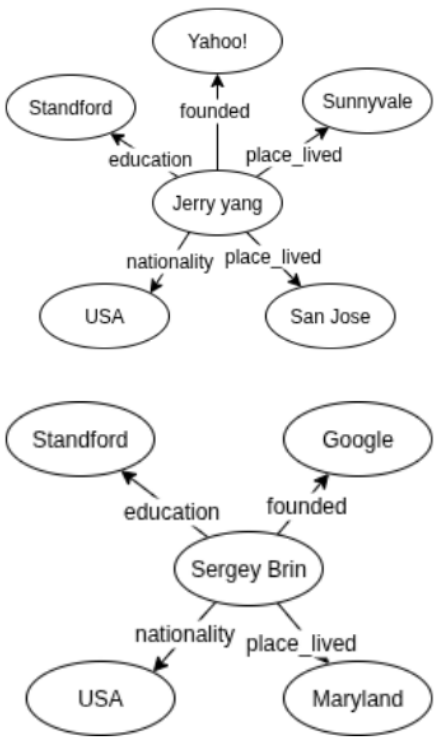
Knowledge Graphs

Structure and Representation

A knowledge Graph



Entity Description

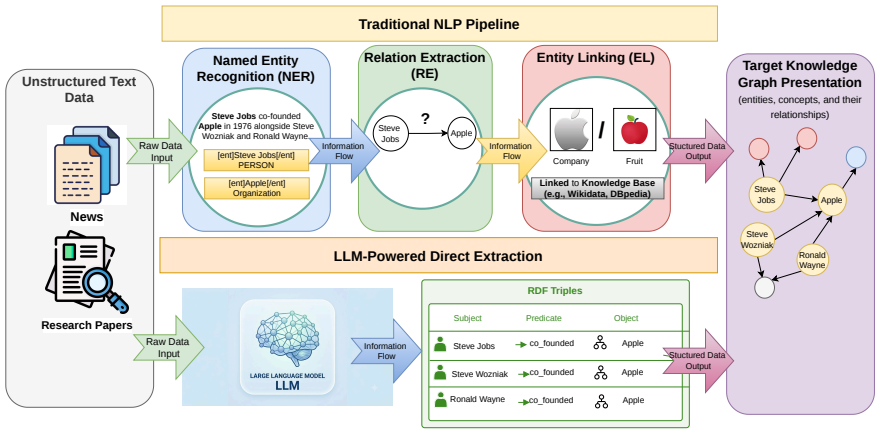


Knowledge Graphs: Construction

Why we need to construct the knowledge graph?

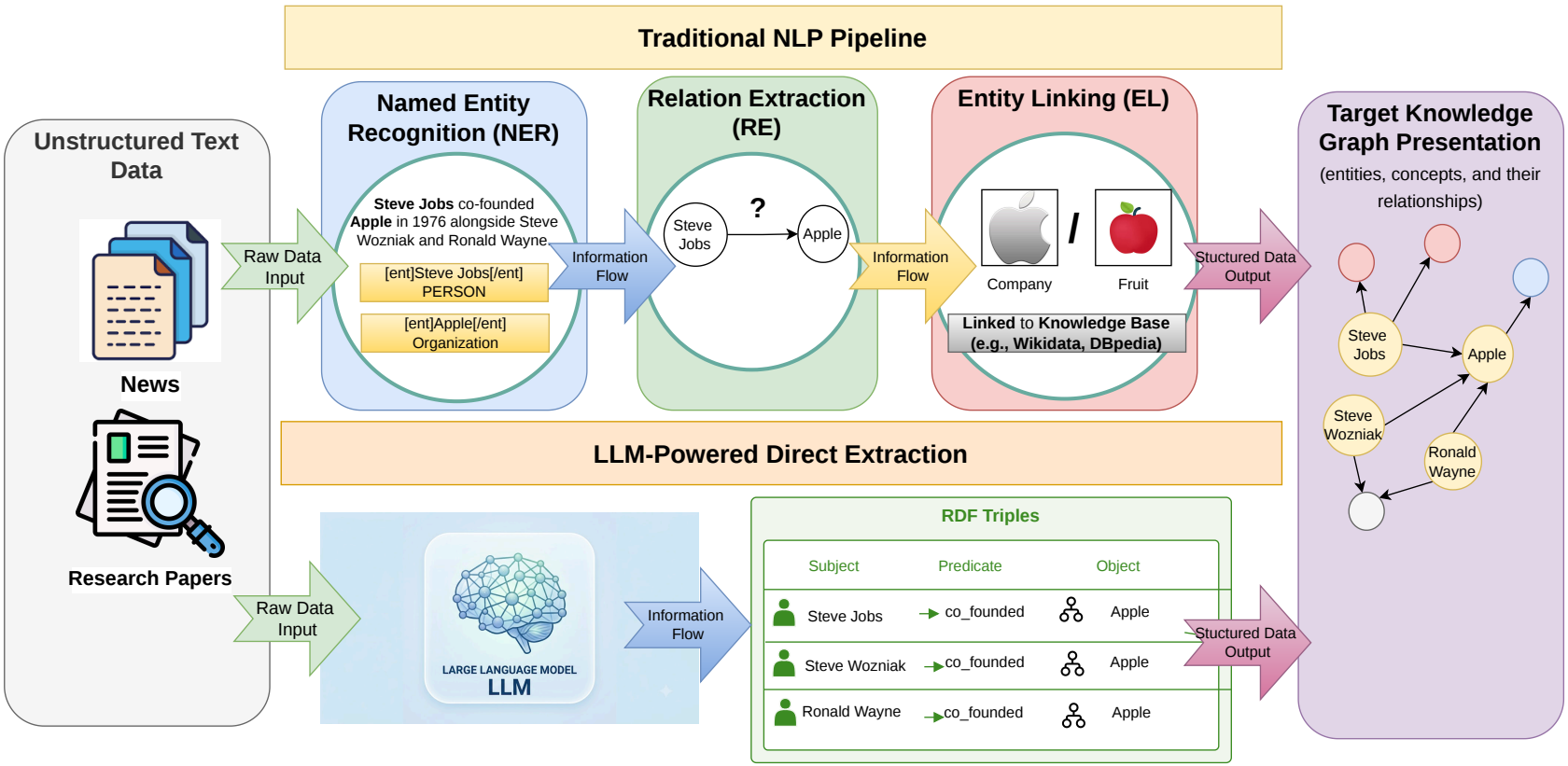
Motivation

- **Problem**: Vast amounts of valuable domain knowledge remain trapped in unstructured raw text.
- **Solution**: Extract structured RDF triples using NLP pipelines (NER, RE, EL) or direct LLMs.
- **Goal**: Build machine-readable, interconnected knowledge graphs from unstructured data sources.



Knowledge Graphs: Construction

Approaches

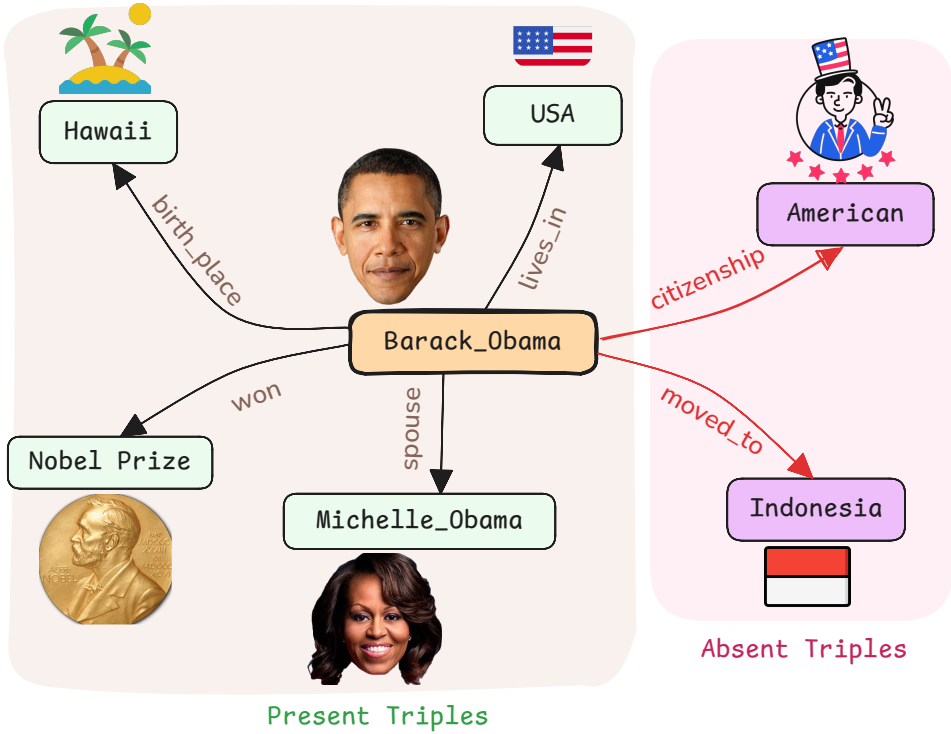


Tiwari, B., Oliviera, I., Firmansyah, A.F., Zahera, H.M., HOPFGARTNER, F., Ngomo, AC.N. (2026). Ontology-Aware Prompting for Knowledge Graph Construction from Text. Accepted at SEMANTICS 2026.

Knowledge Graphs: Reasoning

Motivation

- **Problem**: Incompleteness and missing links in Knowledge Graphs limit downstream tasks.
- **Solution**: Apply logical inference and machine learning (GNNs/Embeddings) for link prediction.
- **Goal**: Infer unstated facts, discover hidden patterns, and complete missing graph relationships.

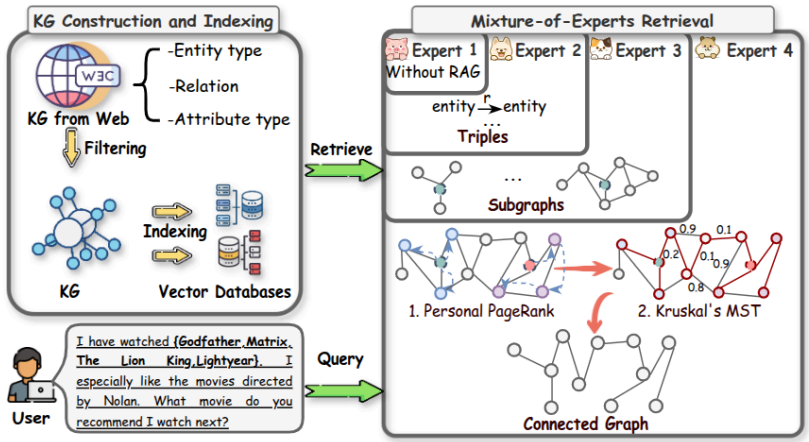


Firmansyah, A.F., Zahera, H.M., Sherif, M.A., Moussallem, D., Ngomo, AC.N. (2025). ANTS: Abstractive Entity Summarization in Knowledge Graphs. The Semantic Web – ESWC 2025. LNCS, vol. 15718, Springer, Cham.

KG Retrieval-Augmented Generation

Motivation

- **Problem**: LLMs suffer from hallucinations, lack domain-specific knowledge, and cannot cite explicit sources.
- **Solution**: Retrieve structured, factual subgraphs from Knowledge Graphs to ground LLM prompts in verifiable context.
- **Goal**: Generate accurate, explainable, and hallucination-free answers enriched with structured semantic knowledge.



Source: https://arxiv.org/pdf/2605.28175

Project Objectives

Addressing the core challenges of Knowledge Graph construction, reasoning, and RAG across the complete research lifecycle.

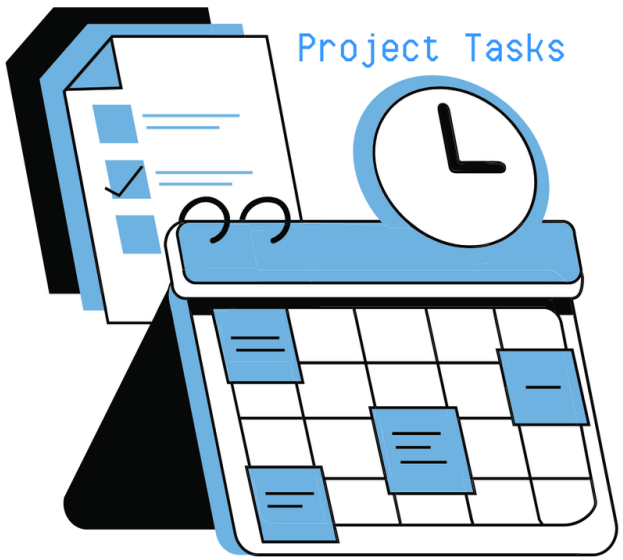
- **Enhance performance:** Improve extraction accuracy, logical inference capability, and factual precision in LLM generation.
- **Explore Advanced Techniques:** Leverage LLMs for automated RDF extraction, Deep Learning for KG reasoning, and KG-RAG architectures.
- **Discover Novel Approaches:** Innovate at the intersection of structured knowledge bases and Generative AI.



Project Tasks

What core tasks will the project group undertake across the research lifecycle?

- **Literature Study:** Survey state-of-the-art methods.
- **Baseline Reproduction:** Implement and test existing baselines on benchmark datasets.
- **Gap Analysis:** Identify limitations in current systems.
- **Method Development and Evaluation:** Design, implement, and evaluate improved pipelines for automated construction, reasoning, or KG-RAG.
- **Graph Visualization**



We Offer

- Training Computer Resources
- Follow-up Thesis Opportunities
- Publication Support



Section 3

Goals & Grading

PG Goals

Current Semester

What are the goals for the current semester?

1. Investigate KG construction, reasoning, and RAG approaches.
2. Run experiments and evaluate each approach.
3. Provide in-depth analysis on what works better and what doesn't.

Grading

How will the students be graded?

- **Achieved Goals:**
  - **Group:** What goals did the group achieve?
  - **Individual:** How much did you contribute towards this goal?
- Reports and Presentations (Group, Individual)
- **Soft Skills (Group, Individual):**
  - Commitment and engagement
  - Collaborative and learning skills
  - Motivation and volitional strategies
  - Scientific writing and self-management

Section 4

Ice Breaker

First Task

"Talk is cheap. Show me the code." – Linus Torvalds

1. Propose at least two papers based on your interests, and select one that is suitable for our topics.
2. Read, analyze, and investigate the content.
3. Present the selected paper in a 10-minute presentation (motivation, problem statement, architecture, experiments: baselines, datasets, setup; results and discussion, conclusion, and key takeaways).

→ **Time Limit:** 2 Week

Next Steps

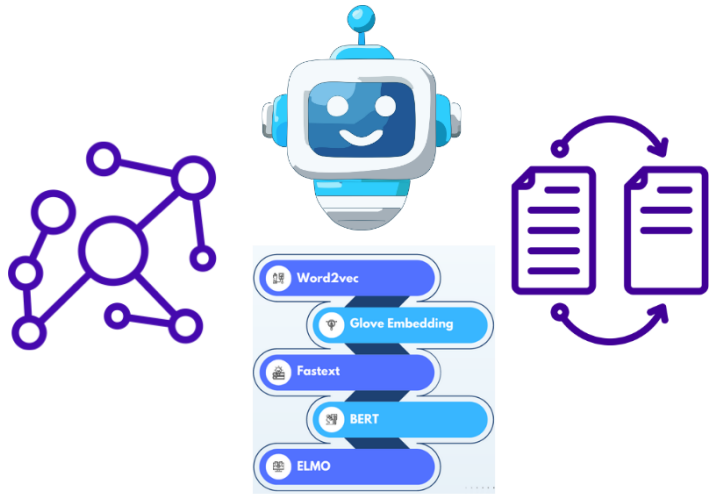
"A place for everything, everything in its place" – Benjamin Franklin

→ Organize Yourselves

- Team Structure: Flat, Hierarchical, Holacratic, …
- Communication: UPB Matrix
- Code Repository: https://github.com/dice-group/KGCR-RAG-2026

→ Try to use open-source tools wherever possible

That's all Folks!



Thank you!

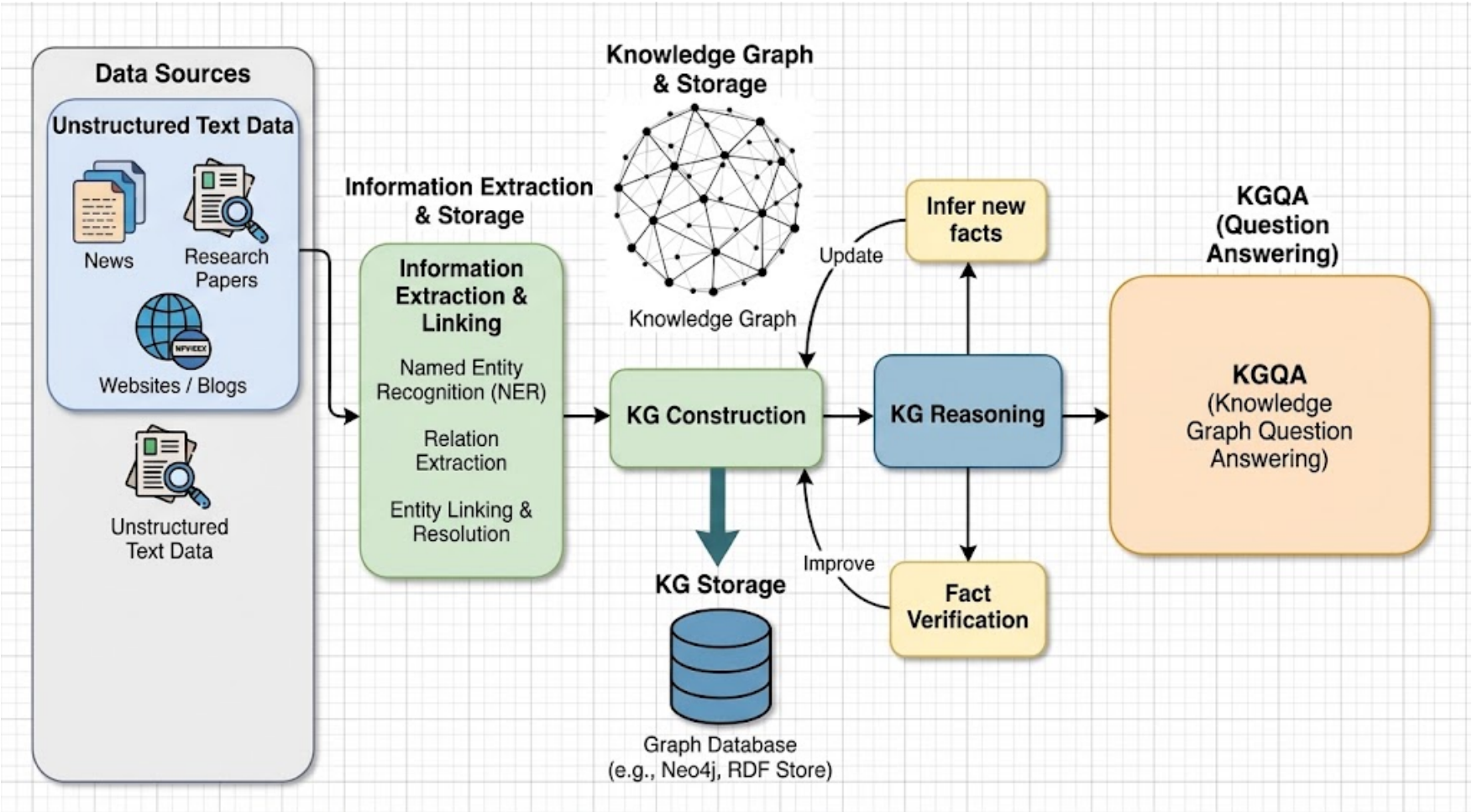
dice-research.org/teaching/KGCR-RAG-2026

Have questions?

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Matrix: @asepff:chat.dice-research.org

All source images are from Google Search and Canva.

Project Group Use Case



Source: Original figure improved by Gemini LLM